Molecular Dynamics Analysis of Substitution Effects on the Thermal Stability of the Trp-Cage Miniprotein

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BIOTECHNOLOGY RESEARCH FELLOWSHIP PROGRAM

INTRODUCTION

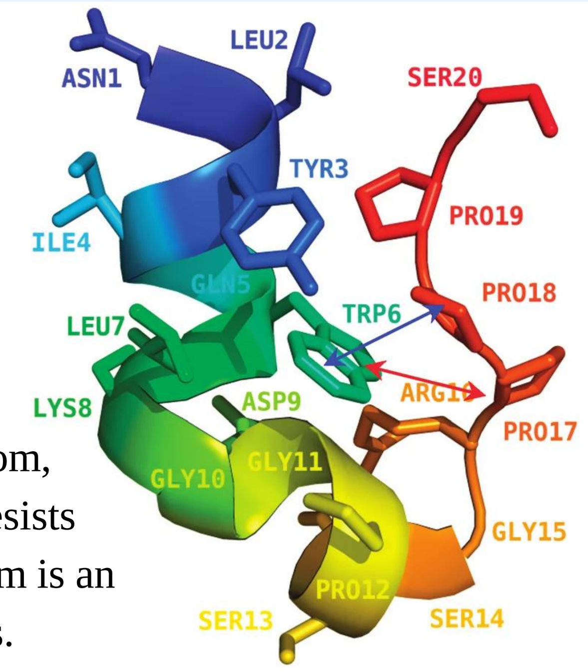


Figure 1. Trp-cage geometry.

- Type 2 diabetes is a rising health concern, accounting for 95% of diabetes cases. One treatment approach targets glucagon-like peptide-1 (GLP-1), a hormone that stimulates insulin secretion but degrades rapidly. Exendin-4, a peptide from Gila monster venom, activates the same receptor but resists degradation, and its synthetic form is an approved therapeutic for diabetes.
- The 20-residue miniprotein TC5b, or **Trp-cage**, was derived from a structural motif in Exendin-4. Consisting of an alpha helix, polyproline II segment, and 3_{10}- helix, it is among the smallest and fastest-folding protein structures known. Because biological activity depends on three-dimensional structure, thermal unfolding causes loss of function. In the Trp-cage, the 3_{10}-helix is the first element to denature and lose its structure at high temperatures, making it a target for stabilization.
- Experimental stability measurements have been reported for mutations at every position in TC5b except its glycines; there is no published data on Gly10 and Gly15, which flank the 3_{10}-helix.
- Because glycine uniquely lacks a side chain and can adopt conformations other residues cannot, substituting it with L-alanine could either stabilize the fold by reducing excessive flexibility or destabilize it by restricting a geometry that the structure requires.

OBJECTIVE

Our objective was to visualize the unfolding of the Trp-cage miniprotein and characterize how replacing Gly10 or Gly15 with L-alanine alters the thermal stability of the Trp-cage miniprotein, using molecular dynamics simulations at 310 K, 360 K, and 410 K.

RESULTS

Figure 3. Protein root mean square deviation (RMSD) over 50 ns.

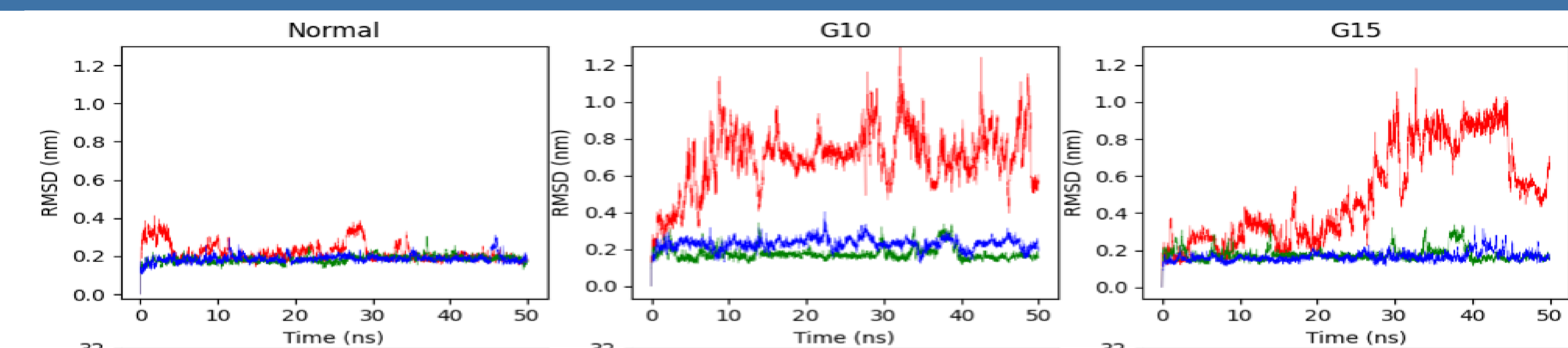


Figure 4. Total solvent accessible surface area (SASA) over 50 ns.

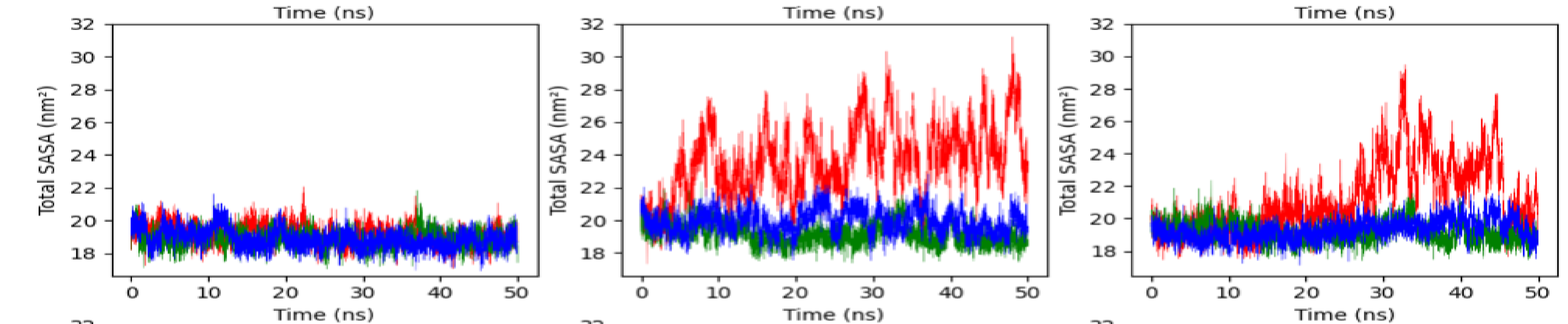


Figure 5. Nonpolar solvent accessible surface area (SASA) over 50 ns, calculated for side-chain atoms of nonpolar residues.

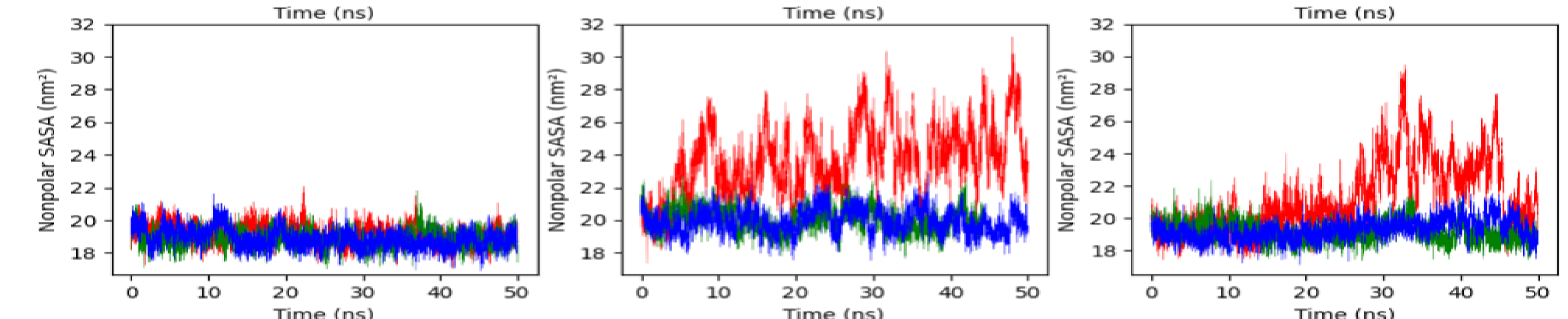


Figure 6. Radius of gyration (Rg) over 50 ns.

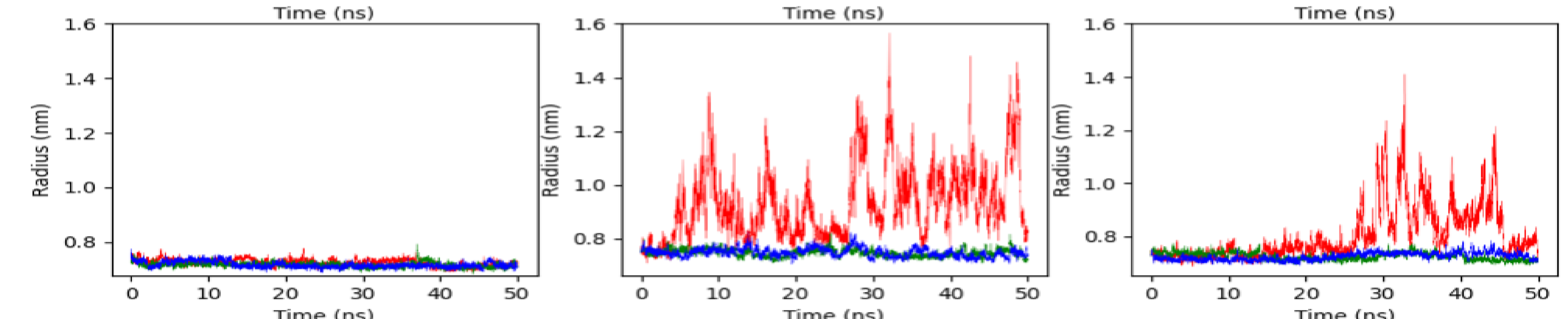
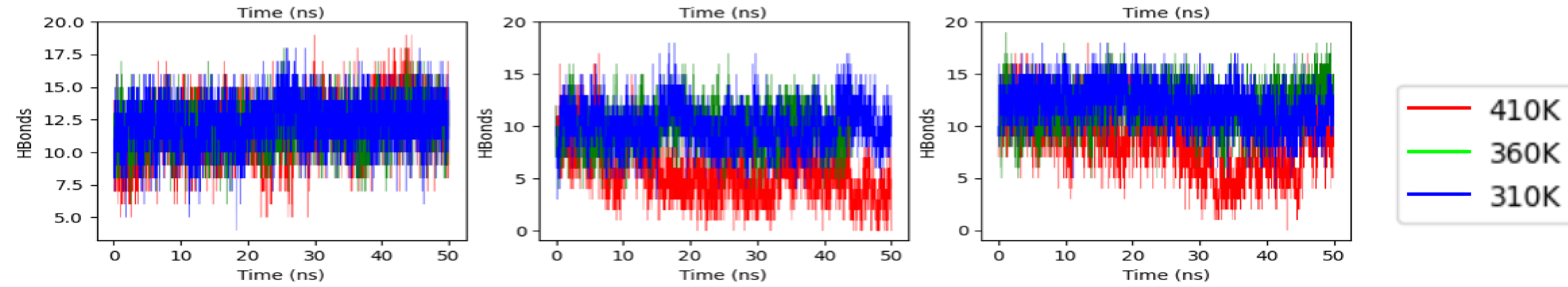


Figure 7. Number of main-chain hydrogen bonds over 50 ns.



METHODOLOGY

- **System Preparation**: The Trp-cage miniprotein TC5b was obtained from the Protein Data Bank (PDB 1L2Y) and used for all simulations. Mutants G10A and G15A were created by replacing the glycine at position 10 or 15 in the TC5b sequence with L-alanine. Each mutant sequence was submitted to AlphaFold.
  Simulation Setup: All simulations were run using GROMACS (2021.4 and 2026.3 versions), using the AMBER99SB-ILDN force field and the SPC/E water model. Each peptide was placed in a cubic box with a minimum solute-box distance of 1.0 nm. Hydrogen atoms present in the input coordinates were discarded and rebuilt by GROMACS. Each system was neutralized and brought to a physiological concentration of 0.15 M NaCl by replacing water molecules with ions.
- **Energy Minimization**: Energy minimization was performed, and potential energy (kJ/mol) was graphed as a function of steps.
- **Equilibration**: NVT and NPT equilibrations were performed, each 100 ps in length with a 2 fs timestep. NVT equilibration was performed to bring the system to the target temperature, and NPT equilibration was performed at 1 bar to establish system density. Temperature (K) and pressure (bar) were graphed as a function of time (ps).
- **Production**: Each production simulation was run for 50 ns (25,000,000 steps and 2 fs timestep) at three temperatures: 310 K, 360 K, and 410 K. 310 K represents physiological temperature and lies below the experimental melting temperature of TC5b (315 K), while 360 K and 410 K lie above it. One simulation was performed per sequence per temperature, and simulations were visualized using Visual Molecular Dynamics.
- **Data analysis**: Root mean square deviation (RMSD), radius of gyration, total and nonpolar solvent accessible surface area (SASA), and hydrogen bond counts were calculated.

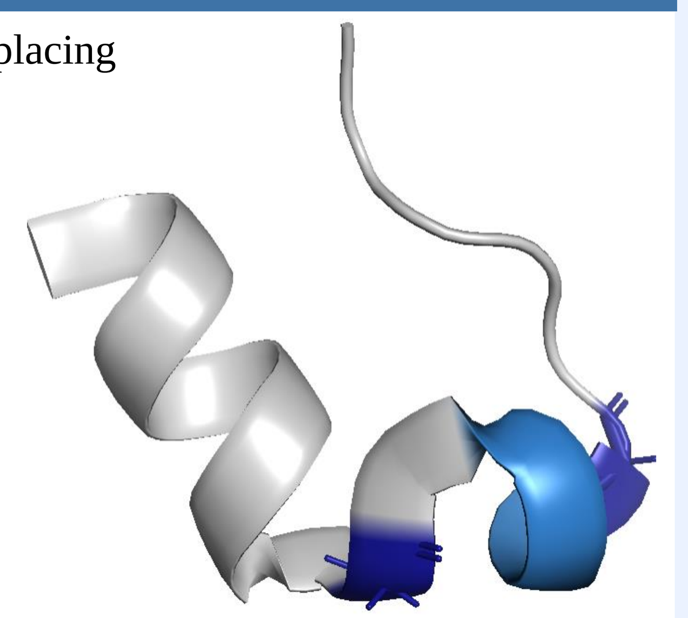


Figure 2. Trp-cage. The 3_{10}-helix is shown in light blue; mutation sites Gly10 and Gly15 are shown in dark blue.

CONCLUSION

- Molecular dynamics simulations of wild-type TC5b and the G10A and G15A mutants at 310, 360, and 410 K show that both substitutions destabilize the Trp-cage fold. All three structures remained folded at 310 and 360 K. At 410 K, both mutants displayed increased root mean square deviation, increased radius of gyration, increased total and nonpolar solvent accessible surface area, and decreased main-chain hydrogen bonds. These changes indicate expansion of the peptide, exposure of buried hydrophobic residues, and loss of backbone secondary structure.
- These results can be attributed to the fact that Gly10 and Gly15 occupied positive-phi conformations with mean values of +115° and +79° (respectively), which is structurally disfavored for L-alanine to occupy because the side-chain clashes with the backbone, and is accessible only to glycine, which lacks a side chain. Substituting L-alanine introduces a residue that cannot adopt the required backbone geometry, inducing strain. Our results are proportional to the magnitude of the phi angle. G10A, which had a phi angle further from the region accessible by L-alanine, showed greater RMSD from the start of the 410 K trajectory, while the RMSD of G15A was similar to the wild-type for about 25 ns.

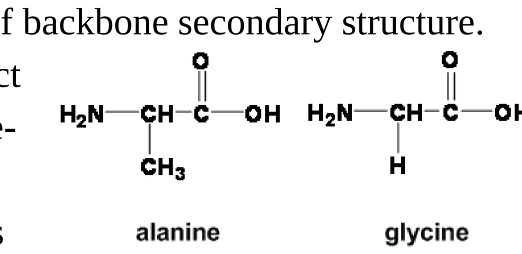


Figure 8. Alanine and glycine.

- Therefore, the conventional expectation that glycine-to-alanine substitution stabilizes a fold does not hold at positions where L-alanine cannot adopt the native backbone conformation. Our results indicate that engineering thermal stability into exendin-4-derived therapeutics cannot proceed through substitution of L-alanine at these positions, and that the flexibility of this loop may be structurally necessary rather than a liability.

LIMITATIONS

- **Lack of several trials**: No replicates were run, so some level of ambiguity exists between the true mutational effects and the run-to-run variation.
- **Short timespan**: Each simulation was run for 50 nanoseconds, which is relatively short relative to other studies.

FURTHER RESEARCH

- **Mapping Gly11**: A mutation at Gly11, part of the hydrophobic core, is the only remaining Tc5b position with no experimental stability measurement.
- **Biological activity**: Future research could conclude whether the destabilization observed in this experiment alters bioactivity.
- **Alternative substitutions:** Because D-alanine can occupy positive phi conformations, substituting glycine with D-alanine may stabilize the fold. Substitutions of different residues other than alanine could also identify whether any side chain is tolerated at these positions.

ACKNOWLEDGEMENTS/REFERENCES

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